EasyToll - Online Toll Payment System

INTERNET PROGRAMMING LAB

AIM

To develop an online toll payment web application using HTML, CSS and JavaScript in which users can register, add money to an eWallet, buy toll eTickets, apply for and renew toll passes and get QR codes; the administrator can create toll details, manage users and toll-gate logins, approve toll passes and generate QR codes; and the toll gate user can scan a QR code, verify the toll pass or ticket and update the toll location, so that drivers can pay in advance and avoid queues.

ALGORITHM

Step 1: Start. Load users, toll gates, passes and tickets from localStorage (seed demo data on the first run).

Step 2: Show the login page with a role selector (User, Admin, Toll Gate User). A new user can register with name, username and password; the username must be unique.

Step 3: Authenticate the username, password and role. On success open the dashboard of that role, otherwise display an error.

Step 4: User - Add Money: validate the amount and add it to the eWallet balance.

Step 5: User - Buy Toll eTicket: select the toll gate, vehicle number and single/return journey; if the wallet balance is at least the fee, deduct the fee, create a ticket ID and generate its QR code; otherwise show "Insufficient balance".

Step 6: User - Apply New Toll Pass: save a pass request with status Pending. User - Renew Toll Pass: deduct the pass fee from the wallet and extend the validity by 30 days (monthly) or 365 days (annual).

Step 7: Admin - Create Toll Details: add a toll gate with its location, single fee and return fee. Manage Toll Login: create a login and assign it to a toll gate. Delete users when required.

Step 8: Admin - Approve Toll Pass: change the status to Approved, set the validity date and generate the QR code (encoded as TOLLPASS|ID|Vehicle|Valid-till) using the QR encoder.

Step 9: Toll Gate User - Scan QR: read the QR text and split it into type, ID and vehicle. For a pass, verify that it exists, is Approved and has not expired. For a ticket, verify that it exists and is unused, then mark it as used.

Step 10: Display VALID (open the barrier) or INVALID with the reason. The toll gate user can also update the toll location of his/her gate.

Step 11: Save every change to localStorage. Logout returns to the login page. Stop.

PROGRAM

Note: script.js uses qr.js, a small open-source QR code encoder (Kazuhiko Arase, MIT licence) placed in the same folder as the other files. It is a third-party library, so its source is not reproduced here.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>EasyToll - Online Toll Payment System</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header><div class="logo">EasyToll <small>Pay in advance. Skip the queue.</small></div>
  <div><span id="who"></span> <button id="logout" class="btn light hidden">Logout</button></div></header>
<main>
<!-- LOGIN / REGISTER -->
<section id="auth" class="card narrow">
  <h2 id="authTitle">Login</h2>
  <label>Login as<select id="role"><option value="user">User</option><option value="admin">Admin</option><option
       value="gate">Toll Gate User</option></select></label>
  <label id="nameRow" class="hidden">Full Name<input id="fullName"></label>
  <label>Username<input id="uname"></label>
  <label>Password<input id="pwd" type="password"></label>
  <button id="authBtn" class="btn">Login</button>
  <a href="#" id="toggleReg">New user? Register here</a>
  <p class="hint">Demo logins: admin / admin123  |  gate1 / gate123  |  arun / user123</p>
  <p id="authMsg" class="msg err"></p>
</section>

<!-- USER -->
<section id="userView" class="hidden">
  <div class="wallet card">eWallet Balance: <b id="bal"></b>
    <input id="addAmt" type="number" placeholder="Amount"><button id="addMoney" class="btn">Add
        Money</button></div>
  <div class="two">
    <div class="card"><h3>Buy Toll eTicket</h3>
      <label>Toll Gate<select id="tGate"></select></label>
      <label>Vehicle Number<input id="tVeh" placeholder="TN01AB1234"></label>
      <label>Journey<select id="tKind"><option value="single">Single</option><option
          value="ret">Return</option></select></label>
      <button id="buyTicket" class="btn">Pay &amp; Get eTicket</button></div>
    <div class="card"><h3>Apply / Renew Toll Pass</h3>
      <label>Vehicle Number<input id="pVeh" placeholder="TN01AB1234"></label>
      <label>Pass Type<select id="pType"><option value="monthly">Monthly - Rs. 1200</option><option
          value="annual">Annual - Rs. 12000</option></select></label>
      <button id="applyPass" class="btn">Apply New Toll Pass</button>
      <label>Renew existing pass<select id="renewSel"></select></label>
      <button id="renewPass" class="btn alt">Renew Toll Pass</button></div>
  </div>
  <p id="userMsg" class="msg"></p>
  <div class="card"><h3>My Toll Passes &amp; eTickets (QR Codes)</h3><div id="myItems"
      class="qrgrid"></div></div>
</section>

<!-- ADMIN -->
<section id="adminView" class="hidden">
  <div class="two">
    <div class="card"><h3>Create Toll Details</h3>
      <label>Toll Gate Name<input id="gName"></label><label>Location<input id="gLoc"></label>
      <label>Single Fee (Rs.)<input id="gSingle" type="number"></label><label>Return Fee (Rs.)<input id="gRet"
          type="number"></label>
      <button id="addGate" class="btn">Add Toll Gate</button></div>
    <div class="card"><h3>Manage Toll Gate Login</h3>
      <label>Gate Username<input id="glUser"></label><label>Password<input id="glPass"></label>
      <label>Assign Gate<select id="glGate"></select></label>
      <button id="addGateLogin" class="btn">Create Login</button></div>
  </div>
  <p id="admMsg" class="msg"></p>
  <div class="card"><h3>Toll Pass Requests (Approve &amp; Generate QR)</h3><div id="passTable"></div></div>
  <div class="card"><h3>Registered Users &amp; Toll Gates</h3><div id="userTable"></div></div>
</section>

<!-- TOLL GATE USER -->
<section id="gateView" class="hidden">
  <div class="two">
    <div class="card"><h3>Scan QR / Verify</h3>
      <p>Scanner input (QR text) - or pick a code to simulate a scan:</p>
      <input id="scanText" placeholder="TOLLPASS|TP1001|TN01AB1234|2026-11-20">
      <select id="simScan"></select>
      <button id="scanBtn" class="btn">Scan &amp; Verify</button>
      <div id="scanResult"></div></div>
    <div class="card"><h3>Update Toll Location</h3>
      <p>Gate: <b id="myGate"></b></p>
      <label>Location<input id="newLoc"></label>
      <button id="updLoc" class="btn">Update Location</button>
```

```
      <p id="gateMsg" class="msg"></p></div>
  </div>
</section>
</main>
<script src="qr.js"></script>
<script src="script.js"></script>
</body>
</html>
```

style.css

```
:root{--navy:#0d3b66;--teal:#12a4a0;--bg:#eef4f8}
*{box-sizing:border-box}
body{margin:0;font-family:"Segoe UI",Arial,sans-serif;background:var(--bg);color:#1c2b36}
header{display:flex;justify-content:space-between;align-
    items:center;background:var(--navy);color:#fff;padding:12px 26px}
.logo{font-size:24px;font-weight:700}.logo small{font-size:12px;font-weight:400;color:#9fd8d6;margin-left:8px}
main{max-width:1050px;margin:18px auto;padding:0 14px}
.card{background:#fff;border-radius:10px;padding:16px;margin-bottom:14px;box-shadow:0 1px 5px
    rgba(13,59,102,.15)}
.narrow{max-width:400px;margin:30px auto}
h2,h3{margin-top:0;color:var(--navy)}
label{display:flex;flex-direction:column;font-size:13px;font-weight:600;margin-bottom:10px;gap:3px}
input,select{padding:8px;border:1px solid #b6c7d4;border-radius:6px;font:inherit}
.btn{background:var(--teal);color:#fff;border:0;border-radius:6px;padding:8px 16px;cursor:pointer;font:inherit}
.btn.alt{background:var(--navy)}.btn.light{background:#fff;color:var(--navy)}.btn.red{background:#c0392b}
.btn.sm{padding:4px 10px;font-size:12px}
.two{display:grid;grid-template-columns:1fr 1fr;gap:14px}
@media(max-width:760px){.two{grid-template-columns:1fr}}
.wallet{display:flex;gap:10px;align-items:center;font-size:18px;flex-wrap:wrap}
.wallet b{color:var(--teal);font-size:24px}
.wallet input{width:130px}
.hint{font-size:12px;color:#557}
.msg{font-weight:700;color:#1e7d3a;min-height:20px}.msg.err{color:#c0392b}
.hidden{display:none!important}
table{width:100%;border-collapse:collapse;font-size:13px}
th,td{padding:7px;border-bottom:1px solid #dde6ee;text-align:left}th{background:#e3eef6}
.qrgrid{display:grid;grid-template-columns:repeat(auto-fill,minmax(190px,1fr));gap:12px}
.qrbox{border:1px dashed var(--teal);border-radius:8px;padding:10px;text-align:center;font-size:12px}
.qrbox canvas{width:120px;height:120px;image-rendering:pixelated}
.badge{padding:2px 8px;border-radius:10px;color:#fff;font-size:11px}
.Approved,.VALID{background:#1e7d3a}.Pending{background:#d98a00}.INVALID{background:#c0392b}
.result{margin-top:12px;padding:12px;border-radius:8px;font-weight:700}
.result.ok{background:#e1f6e8;color:#1e7d3a}.result.bad{background:#fde7e4;color:#c0392b}
#scanText{width:100%;margin-bottom:8px}#simScan{width:100%;margin-bottom:8px}
```

script.js

```
const $ = id => document.getElementById(id);
const FEES = {monthly:1200, annual:12000};
const today = () => new Date().toISOString().slice(0,10);
const plus = (d, n) => { const x = new Date(d); x.setDate(x.getDate()+n); return x.toISOString().slice(0,10); };
const seed = () => ({
  users:[{u:'admin',p:'admin123',role:'admin',name:'Administrator'},
         {u:'gate1',p:'gate123',role:'gate',name:'Gate Operator',gate:1},
         {u:'arun',p:'user123',role:'user',name:'Arun Kumar',wallet:2500}],
  gates:[{id:1,name:'Sriperumbudur Toll Plaza',loc:'NH-48, Km 36',single:85,ret:125},
         {id:2,name:'Walajapet Toll Plaza',loc:'NH-48, Km 104',single:110,ret:165}],

      passes:[{id:'TP1001',user:'arun',vehicle:'TN09AB1234',type:'monthly',status:'Approved',valid:plus(today(),
      20)},
          {id:'TP1002',user:'arun',vehicle:'TN09CD5678',type:'annual',status:'Pending',valid:''}],
  tickets:[], nextPass:1003, nextTicket:5001, nextGate:3});
let db = JSON.parse(localStorage.getItem('easytoll') || 'null') || seed();
const save = () => localStorage.setItem('easytoll', JSON.stringify(db));
let me = null, regMode = false;

/* ---------- QR ---------- */
function drawQR(canvas, text){
  const q = new QRCodeLib(-1, 1); q.addData(text); q.make();
  const n = q.getModuleCount(), s = 4, pad = 4, size = (n + 2*pad) * s;
  canvas.width = canvas.height = size;
  const c = canvas.getContext('2d'); c.fillStyle = '#fff'; c.fillRect(0,0,size,size); c.fillStyle = '#000';
  for(let r=0;r<n;r++) for(let k=0;k<n;k++) if(q.isDark(r,k)) c.fillRect((k+pad)*s,(r+pad)*s,s,s);
}
const passQR = p => `TOLLPASS|${p.id}|${p.vehicle}|${p.valid}`;
const tickQR = t => `TOLLTICKET|${t.id}|${t.vehicle}|${t.date}`;

/* ---------- login / register ---------- */
$('toggleReg').onclick = e => { e.preventDefault(); regMode = !regMode;
  $('authTitle').textContent = regMode ? 'User Registration' : 'Login';
  $('authBtn').textContent = regMode ? 'Register' : 'Login';
  $('nameRow').classList.toggle('hidden', !regMode);
  $('toggleReg').textContent = regMode ? 'Back to Login' : 'New user? Register here'; };
$('authBtn').onclick = () => {
  const u = $('uname').value.trim(), p = $('pwd').value;
  if(regMode){
    if(!u || !p || !$('fullName').value) return err('Fill all fields.');
    if(db.users.some(x => x.u === u)) return err('Username already exists.');
    db.users.push({u, p, role:'user', name:$('fullName').value, wallet:0}); save();
    regMode = true; $('toggleReg').click(); return err('Registered! Please login.', true);
  }
  const usr = db.users.find(x => x.u === u && x.p === p && x.role === $('role').value);
  if(!usr) return err('Invalid username, password or role.');
  me = usr; render();
};
function err(t, ok){ $('authMsg').textContent = t; $('authMsg').className = 'msg' + (ok ? '' : ' err'); }
$('logout').onclick = () => { me = null; $('pwd').value = ''; render(); };
function say(el, t, bad){ $(el).textContent = t; $(el).className = 'msg' + (bad ? ' err' : ''); }

/* ---------- user ---------- */
const gateName = id => db.gates.find(g => g.id == id).name;
$('addMoney').onclick = () => {
  const a = +$('addAmt').value; if(a <= 0) return say('userMsg','Enter a valid amount.',true);
  me.wallet += a; save(); $('addAmt').value = ''; say('userMsg', `Rs. ${a} added to eWallet.`); renderUser();
};
$('buyTicket').onclick = () => {
  const g = db.gates.find(x => x.id == $('tGate').value), veh = $('tVeh').value.trim().toUpperCase();
  if(!veh) return say('userMsg','Enter the vehicle number.',true);
  const fee = $('tKind').value === 'ret' ? g.ret : g.single;
  if(me.wallet < fee) return say('userMsg',`Insufficient balance. Fee is Rs. ${fee}.`,true);
  me.wallet -= fee;
  const t = {id:'TT' + db.nextTicket++, user:me.u, gate:g.id, vehicle:veh, kind:$('tKind').value, fee,
      date:today(), used:false};
  db.tickets.push(t); save(); say('userMsg', `eTicket ${t.id} purchased for Rs. ${fee}. Show the QR code at the
      toll gate.`); renderUser();
};
$('applyPass').onclick = () => {
  const veh = $('pVeh').value.trim().toUpperCase(); if(!veh) return say('userMsg','Enter the vehicle
      number.',true);
  const p = {id:'TP' + db.nextPass++, user:me.u, vehicle:veh, type:$('pType').value, status:'Pending',
      valid:''};
  db.passes.push(p); save(); say('userMsg', `Pass request ${p.id} submitted. Waiting for admin approval.`);
      renderUser();
};
$('renewPass').onclick = () => {
  const p = db.passes.find(x => x.id === $('renewSel').value); if(!p) return say('userMsg','No approved pass to
      renew.',true);
  if(me.wallet < FEES[p.type]) return say('userMsg','Insufficient eWallet balance for renewal.',true);
  me.wallet -= FEES[p.type]; const base = p.valid > today() ? p.valid : today();
```

```
  p.valid = plus(base, p.type === 'monthly' ? 30 : 365); save();
  say('userMsg', `Pass ${p.id} renewed. New validity: ${p.valid}.`); renderUser();
};
function qrBox(title, sub, payload){
  const d = document.createElement('div'); d.className = 'qrbox';
  d.innerHTML = `<b>${title}</b><br>${sub}<br>`; const c = document.createElement('canvas'); drawQR(c, payload);
       d.appendChild(c);
  return d;
}
function renderUser(){
  $('bal').textContent = 'Rs. ' + me.wallet;
  $('tGate').innerHTML = db.gates.map(g => `<option value="${g.id}">${g.name} (Rs. ${g.single} /
      ${g.ret})</option>`).join('');
  const mine = db.passes.filter(p => p.user === me.u), appr = mine.filter(p => p.status === 'Approved');
  $('renewSel').innerHTML = appr.map(p => `<option value="${p.id}">${p.id} - ${p.vehicle} (valid
      ${p.valid})</option>`).join('');
  const box = $('myItems'); box.innerHTML = '';
  mine.forEach(p => { if(p.status === 'Approved') box.appendChild(qrBox('Toll Pass ' + p.id,
      `${p.vehicle}<br>Valid till ${p.valid}`, passQR(p)));
    else { const d = document.createElement('div'); d.className = 'qrbox'; d.innerHTML = `<b>Toll Pass
        ${p.id}</b><br>${p.vehicle}<br><span class="badge Pending">Pending approval</span>`; box.appendChild(d);
         } });
  db.tickets.filter(t => t.user === me.u).forEach(t => box.appendChild(qrBox('eTicket ' + t.id,
      `${t.vehicle}<br>${gateName(t.gate)}<br>Rs. ${t.fee} (${t.kind})`, tickQR(t))));
}

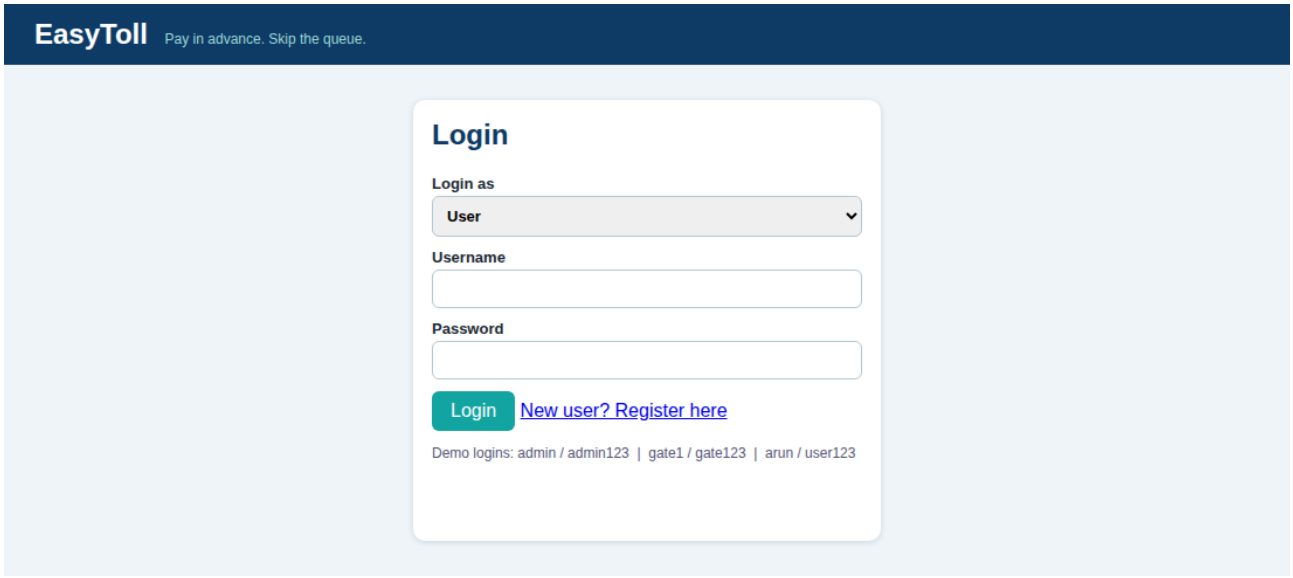
/* ---------- admin ---------- */
$('addGate').onclick = () => {
  if(!$('gName').value || !+$('gSingle').value) return say('admMsg','Enter gate name and fees.',true);
  db.gates.push({id:db.nextGate++, name:$('gName').value, loc:$('gLoc').value, single:+$('gSingle').value,
      ret:+$('gRet').value || 0}); save();
  say('admMsg','Toll gate created.'); renderAdmin();
};
$('addGateLogin').onclick = () => {
  const u = $('glUser').value.trim(); if(!u || !$('glPass').value) return say('admMsg','Enter username and
      password.',true);
  db.users.push({u, p:$('glPass').value, role:'gate', name:'Gate Operator', gate:+$('glGate').value}); save();
      say('admMsg','Toll gate login created.'); renderAdmin();
};
function approve(id){ const p = db.passes.find(x => x.id === id); p.status = 'Approved'; p.valid = plus(today(),
     p.type === 'monthly' ? 30 : 365); save(); renderAdmin(); }
function delUser(u){ db.users = db.users.filter(x => x.u !== u); save(); renderAdmin(); }
function renderAdmin(){
  $('glGate').innerHTML = db.gates.map(g => `<option value="${g.id}">${g.name}</option>`).join('');
  $('passTable').innerHTML = '<table><tr><th>Pass
      ID</th><th>User</th><th>Vehicle</th><th>Type</th><th>Status</th><th>Valid Till</th><th>Action</th></tr>' +
    db.passes.map(p => `<tr><td>${p.id}</td><td>${p.user}</td><td>${p.vehicle}</td><td>${p.type}</td><td><span
        class="badge ${p.status}">${p.status}</span></td><td>${p.valid || '-'}</td>
    <td>${p.status === 'Pending' ? `<button class="btn sm" onclick="approve('${p.id}')">Approve &amp;amp; Generate
        QR</button>` : 'QR generated'}</td></tr>`).join('') + '</table>';
  $('userTable').innerHTML =
      '<table><tr><th>Username</th><th>Name</th><th>Role</th><th>Details</th><th>Action</th></tr>' +
    db.users.filter(u => u.role !== 'admin').map(u => `<tr><td>${u.u}</td><td>${u.name}</td><td>${u.role}</td>
    <td>${u.role === 'user' ? 'Wallet Rs. ' + u.wallet : gateName(u.gate)}</td><td><button class="btn sm red"
        onclick="delUser('${u.u}')">Delete</button></td></tr>`).join('') +
    '</table><br><table><tr><th>Gate</th><th>Location</th><th>Single</th><th>Return</th></tr>' +
    db.gates.map(g => `<tr><td>${g.name}</td><td>${g.loc}</td><td>Rs. ${g.single}</td><td>Rs.
        ${g.ret}</td></tr>`).join('') + '</table>';
}

/* ---------- toll gate user ---------- */
function renderGate(){
  const g = db.gates.find(x => x.id === me.gate);
  $('myGate').textContent = g.name + ' - ' + g.loc; $('newLoc').value = g.loc;
  const opts = db.passes.filter(p => p.status === 'Approved').map(p => passQR(p)).concat(db.tickets.map(t =>
      tickQR(t)));
  opts.push('TOLLPASS|TP9999|TN00XX0000|2026-01-01');
  $('simScan').innerHTML = '<option value="">-- simulate scan --</option>' + opts.map(o =>
      `<option>${o}</option>`).join('');
}
$('simScan').onchange = () => { $('scanText').value = $('simScan').value; };
$('updLoc').onclick = () => { db.gates.find(x => x.id === me.gate).loc = $('newLoc').value; save();
    say('gateMsg','Toll location updated.'); renderGate(); };
$('scanBtn').onclick = () => {
  const [kind, id, veh, extra] = $('scanText').value.trim().split('|'); let ok = false, text = 'Unknown QR code
      - INVALID';
  if(kind === 'TOLLPASS'){
    const p = db.passes.find(x => x.id === id);
    if(p && p.status === 'Approved' && p.valid >= today()){ ok = true; text = `VALID TOLL PASS ${id} - Vehicle
        ${p.vehicle} - valid till ${p.valid}. Open the barrier.`; }
    else text = p ? `INVALID - pass ${id} is ${p.status === 'Approved' ? 'expired' : 'not approved'}.` :
        `INVALID - pass ${id} not found.`;
  } else if(kind === 'TOLLTICKET'){
    const t = db.tickets.find(x => x.id === id);
```

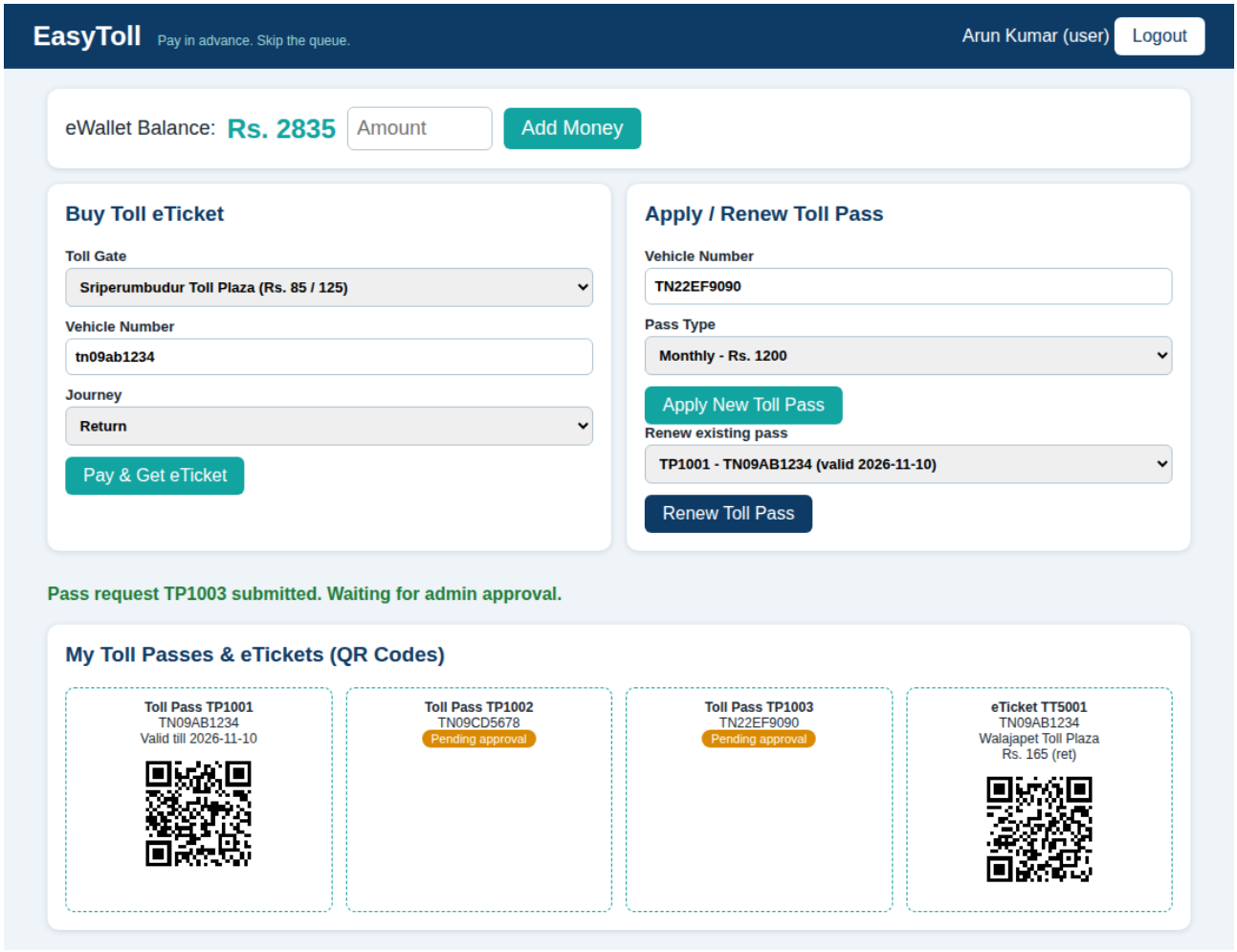
```
    if(t && !t.used){ t.used = true; save(); ok = true; text = `VALID eTICKET ${id} - Vehicle ${t.vehicle} - Rs.
         ${t.fee} paid. Ticket marked as used.`; }
    else text = t ? `INVALID - eTicket ${id} was already used.` : `INVALID - eTicket ${id} not found.`;
  }
  $('scanResult').innerHTML = `<div class="result ${ok ? 'ok' : 'bad'}">${text}</div>`;
};

/* ---------- screen control ---------- */
function render(){
  ['auth','userView','adminView','gateView'].forEach(s => $(s).classList.add('hidden'));
  $('logout').classList.toggle('hidden', !me); $('who').textContent = me ? `${me.name} (${me.role})` : '';
  if(!me) return $('auth').classList.remove('hidden');
  if(me.role === 'user'){ $('userView').classList.remove('hidden'); renderUser(); }
  if(me.role === 'admin'){ $('adminView').classList.remove('hidden'); renderAdmin(); }
  if(me.role === 'gate'){ $('gateView').classList.remove('hidden'); renderGate(); $('scanResult').innerHTML =
      ''; }
}
render();
```

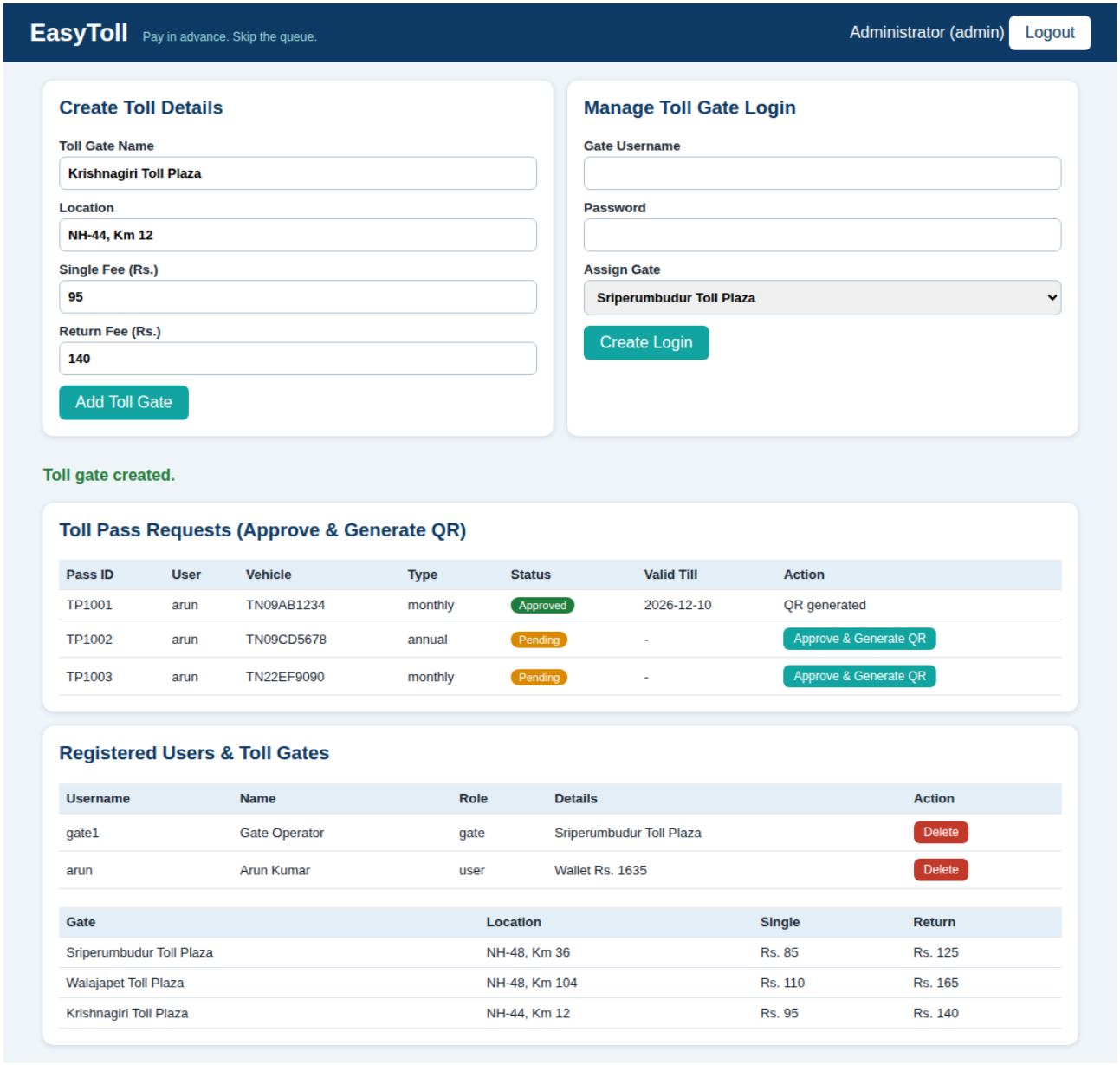
OUTPUT



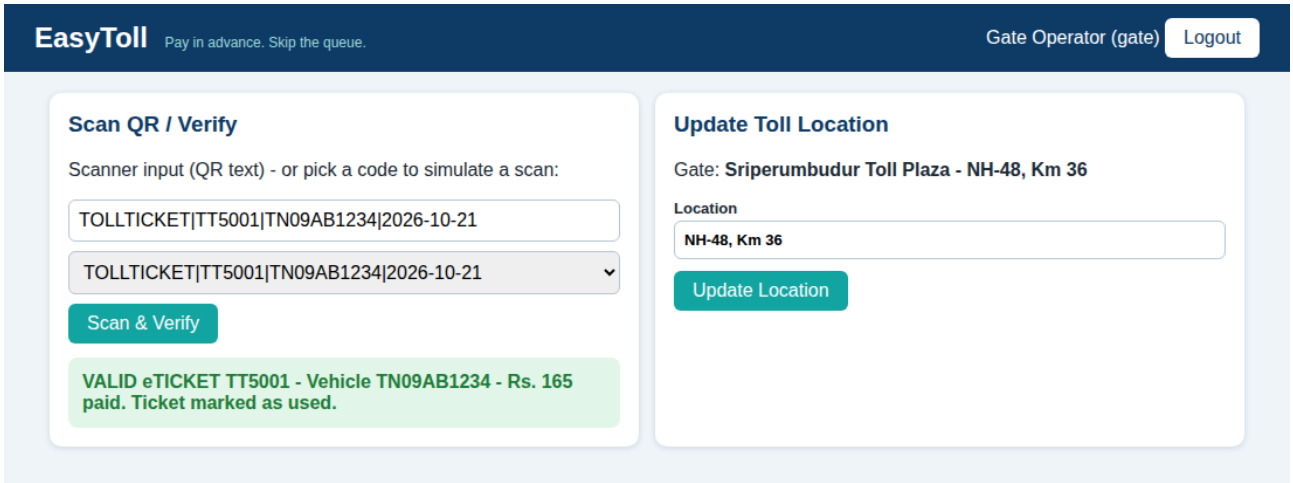
Login page with role selection (User / Admin / Toll Gate User) and new-user registration link



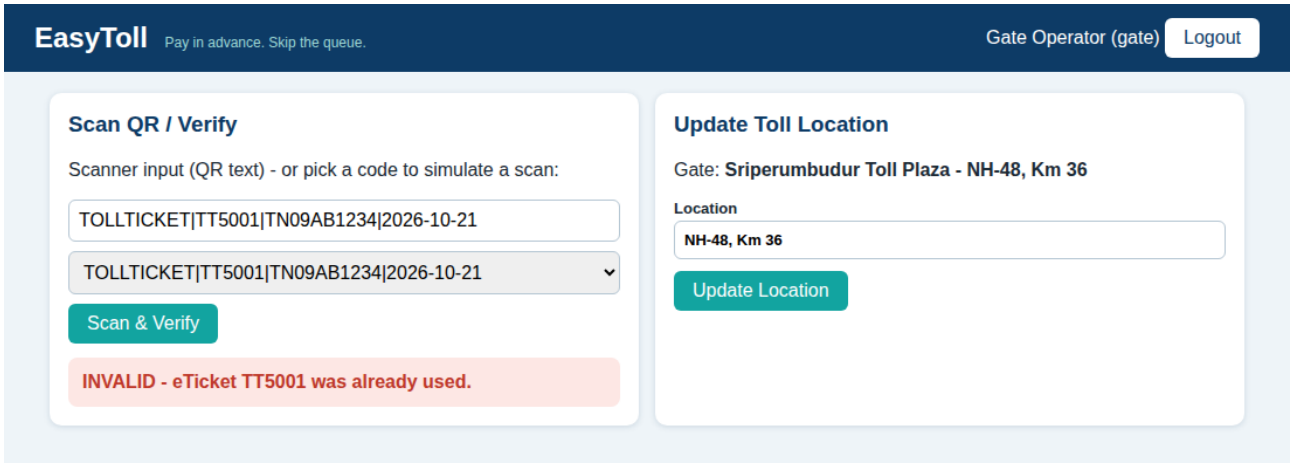
User dashboard: eWallet top-up, eTicket purchase, new pass application and QR codes for the toll pass and eTicket



Admin dashboard: create toll details, approve toll pass requests, generate QR codes and manage users



Toll gate user: scanning the QR code and verifying the toll eTicket (valid)



Re-scanning the same eTicket is rejected as already used

RESULT

The EasyToll online toll payment system was developed using HTML, CSS and JavaScript. Users could add money, buy eTickets, apply for and renew toll passes and obtain QR codes; the admin could create toll details, approve passes and generate QR codes; and the toll gate user could scan and verify passes and tickets. The output was verified successfully.